

Physical Geography

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I. Question 1-12: Multiple choice(s)/true-false and short answer questions (5% each; no partial credits). THIS ASSIGNMENT SHEET IS DOUBLE-SIDED.

1. The major difference between tropical rainforest and tropical monsoon (dry) forest is the amount of annual precipitation (a) True (b) False
a

2. Order from dry to wet climates:

(a) Savanna grassland (b) tropical scrub (c) tropical desert (d) equatorial rainforest (e) savanna woodland

c b a e d

3. Liana is a woody species commonly found in tropical forests (a) True (b) False

a

4. What are the main plant physiological symptoms resulting in massive tree mortality in past 20 years (a) hydraulic failure (b) nutrient shortage (c) xylem decomposition (d) photosynthesis failure (e) carbon starvation

a d e

5. What are the main threats of tropical montane cloud forests? (a) cattle ranching (b) agriculture (c) anthropogenic development (d) deforestation (e) flood

a b c d

6. Give a real-world example that climate change (drought) in concert with land use resulting in the shrinkage of inland water body:

Aral sea, 農業用水

7. What can be commonly found in lowland evergreen rainforests? (a) dense tree canopies (b) tall trees (c) mosses (d) gnarled trees (e) buttresses

a b e

8. According to the short documentary film "Snows of the Nile", please provide ONE evidence to show the retreat of glaciers.

對照歷史照片與現地重拍:冰河末端明顯後退、面積/厚度縮小,裸岩增多。

9. Can we find tropical dry forests in Taiwan?

不可以

10. What are the crucial roles of tropical montane cloud forests? (a) High quality water supplies

(b) Dry season water supplies (c) Unique biodiversity (d) High carbon storages (e) Small areas

a b c d

11. What is the telescoping effect?

小山或孤立山的植被垂直分帶被壓縮,使本來較高海拔才出現的森林型(如苔蘚矮林/上部山地特徵)在較低海拔就出現

12. What are the major high mountain tea harvesting seasons in Taiwan? (a) Spring (b) Summer (c) Autumn (d) Winter (e) All year round

a d

II. Short essay

13. Please briefly explain why elevated air temperature may shrink the cloud band? (15%)

氣溫上升 → 飽和水氣壓增加,若水氣供給沒同步增加,則相對濕度下降。

相對濕度下降 → 空氣要抬升到更高高度才會達到飽和凝結,LCL(lifted condensation level)上升、雲底抬升。

同時升溫增強 蒸散(ET)、加大乾季水分壓力,使可形成持續雲霧的高度範圍變窄。

結果:原本穩定存在於某高程帶的雲霧/雲帶,可能上移、變薄或出現頻率下降("cloudstripping/水平降水"貢獻減少)

14. Here is the centurial climate anomaly data from the Alishan weather station with tea production of the Greater Alishan region. I have spent a lengthy amount of time explaining this figure in the class. Therefore, I will not explain the figure here to save some space. Here are the questions:

14.1. What is an "anomaly"? (5%)

某一年（或某一時期）的觀測值與該地長期平均值之間的差異。

使用距平可以消除不同年份之間絕對數值大小的影響，讓我們更清楚比較氣候變化的趨勢與年際變動。

14.2. Why we need an 11-year average moving window to smooth the annual precipitation anomaly and what is the pattern? (10%)

使用 11 年移動平均 來平滑年降雨距平，可以降低短期雜訊，由圖中可見，經過平滑後的降雨距平呈現長期乾濕交替的波動型態，里山地區降雨具有明顯的多年代變動特性，近年降雨變異幅度亦有增大的現象。

14.3. Please describe the trend of tea production through years. What might cause the fluctuation? (10%)

茶葉產量整體呈現長期上升趨勢，尤其在近數十年間增幅明顯，但年際之間仍存在顯著波動。

茶葉產量波動的可能原因包括：

氣候因素，如降雨距平、乾旱、颱風與極端降雨事件，會影響茶樹生長與採收時程；

溫度變化，升溫可能延長生長季，但極端高溫亦可能造成生理壓力；

人為因素，如茶園擴張、品種改良、管理技術提升與市場需求變化；